

# Project Checkpoint 02

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## Mark of Ascension

*Game Development with Unity*

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|------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>Team Name</b> | Ascension Studio                                                                                                |
| <b>Members</b>   | Aarush Patel, 2330241, 2330241@champlaincollege.qc.ca<br>Elias Larhdaf, 2231248, 2231248@champlaincollege.qc.ca |
| <b>Genre</b>     | 2D top-down action game with dungeon-style combat and progression                                               |
| <b>Platform</b>  | PC Windows                                                                                                      |
| <b>GitHub</b>    | <a href="https://github.com/AarushP06/Mark-of-Ascension">https://github.com/AarushP06/Mark-of-Ascension</a>     |

## Project Summary

Mark of Ascension is a 2D top-down action game where the player controls Kai, a weak hunter who enters dangerous gates and becomes stronger by defeating enemies and bosses. The game focuses on movement, combat, progression, and level-based gameplay. For this checkpoint, our team focused on building the core playable foundation of the game. This includes the main menu, lobby flow, first playable stage, movement between scenes, camera setup, stage layout, physics-based wall collision, and placeholder gameplay systems for hazards and portals.

### 1. Nine Features or Work Steps

| No. | Feature / Work Step                                                 |
|-----|---------------------------------------------------------------------|
| 1   | Main menu scene with Start and Exit buttons                         |
| 2   | Lobby scene where the player begins after starting the game         |
| 3   | Scene transition system between menu, lobby, and Stage 1            |
| 4   | Persistent player handling across scenes                            |
| 5   | Stage 1 scene creation and map layout                               |
| 6   | Tilemap system using Ground, Walls, Decor, and Hazards              |
| 7   | Character movement system                                           |
| 8   | Physics and collision for walls and map boundaries                  |
| 9   | Prefab and placeholder gameplay objects such as portals and hazards |

### 2. Four Implemented Features for Checkpoint 02

#### Feature 1: Main Menu Scene

We created a MainMenu scene for the game. The menu includes a title, a Start button, and an Exit button. The Start button loads the lobby scene, and the Exit button safely closes the game in a build and stops play mode in the Unity editor.

**Why we chose this feature:** This feature was important because every game needs a starting point. It also shows that our scene flow begins in a complete and organized way.

#### Feature 2: Lobby Scene and Portal Flow

We connected the existing lobby scene to the first stage. The player starts in the lobby, and when they enter the gate portal, the game loads Stage01. This shows the beginning of the real game flow.

**Why we chose this feature:** We wanted to prove that the game can move from a safe area into the first playable level. This is a core part of the player experience.

#### Feature 3: Stage 1 Scene and Level Layout

We created the first stage, Stage01, and built its prototype layout. The stage includes an entry area, a main combat room, a corridor, and a boss room. The scene uses four tilemap layers: Ground, Walls, Decor, and Hazards. The scene also includes markers such as PlayerSpawn, EnemySpawn\_01, EnemySpawn\_02, BossSpawn, ExitPortal\_Closed, and BossGatePlaceholder.

**Why we chose this feature:** This feature was one of the most important because it gives us the first real level of the game. It also demonstrates scene design, map building, and level structure.

### Feature 4: Physics, Camera, and Placeholder Gameplay Systems

We added wall blockers, map bounds, hazard triggers, camera improvements, and placeholder stage flow systems. The stage includes poison hazard trigger objects, a boss gate placeholder, an exit portal placeholder, and a stage flow controller. We also improved the camera by adjusting the lobby zoom and adding a follow camera for Stage01.

**Why we chose this feature:** We chose this feature because it helps make the prototype playable. Even though the combat and boss are still placeholders, the player can move through the environment properly, and the scene feels like a working game prototype.

### 3. Current Project Progress

#### Completed So Far

- Main menu scene and UI
- Start and Exit button logic
- Build scene list updated
- Lobby connected to Stage01
- Persistent player system
- Stage01 scene and map layout
- Tilemap layers and markers
- Hazard, gate, and exit placeholders
- Return portal from Stage01 to lobby
- Camera improvements and map bounds

#### Scripts Added

- MainMenuController.cs
- PoisonHazardTrigger.cs
- ExitPortalPlaceholder.cs
- Stage01FlowController.cs
- ScenePortal.cs
- SceneTransitionState.cs
- PersistentPlayer.cs
- StagePlayerBootstrap.cs
- CameraFollow2D.cs

Current prototype state: The player can start from the menu, enter the lobby, use the lobby portal to enter Stage01, move through Stage01, and return to the lobby. The boss flow is still placeholder-based.

### 4. Work Distribution

**Aarush Patel:** Main menu setup, scene flow, lobby-to-Stage01 transition, Stage01 layout and hierarchy, camera adjustments, hazard and portal placeholders, and checkpoint documentation.

**Elias Larhdaf:** Asset integration support, map decoration support, testing and feedback on scene flow, stage structure review, and support for future enemy and boss planning.

### 5. Next 4 Weeks Plan

| Week   | Planned Work                                                                                                                              |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1 | Add player health system, damage handling, poison hazard damage, and a simple HUD with health display.                                    |
| Week 2 | Create a simple enemy prefab, add enemy movement or chase AI, add enemy health and damage, and connect enemy spawns in Stage01.           |
| Week 3 | Create a boss placeholder enemy, add boss health, add a basic boss attack pattern, and add a boss defeat condition.                       |
| Week 4 | Unlock the exit portal after boss defeat, improve Stage01 visuals, add sound effects and ambience, and prepare Stage02 or a victory flow. |

### 6. What Still Needs to Be Done

- Real combat system
- Enemy AI, enemy health, and enemy damage

- Player attack system
- Real boss system and boss defeat logic
- Real exit portal unlocking
- Player health, damage, death, and respawn
- HUD and stage objective text
- Audio and atmosphere polish
- Better sewer decoration and portal visuals
- Cleaner future scene flow

## **8. Conclusion**

For Checkpoint 02, our team focused on building a playable foundation for Mark of Ascension. We successfully implemented the menu, lobby flow, first stage, camera system, physics boundaries, tilemap structure, and placeholder progression objects. These features were chosen because they establish the core structure of the game and allow us to demonstrate real progress. The next phase of development will focus on combat, enemies, boss mechanics, UI, and progression systems so the game becomes a complete playable experience.